

The Poisson Distribution

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Probability mass function (PMF) ([Note 241](#)); Expected value and variance of a random variable ([Note 332](#)).
- **Compare with:** Bernoulli and binomial distributions ([Note 270](#)).

1. Overview

Key point: The Poisson distribution gives the probability of each count of events (0, 1, 2, ...) in a fixed interval of time or space, when we know only the average count λ .

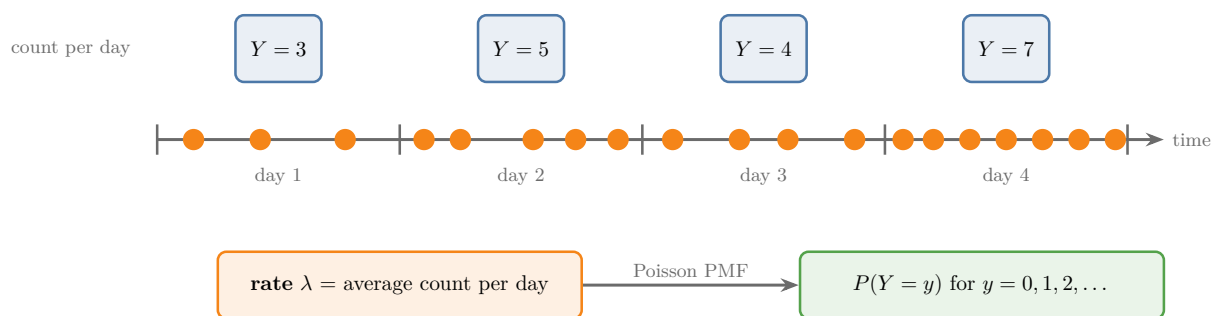


Figure 1: Events on a time line, counted per day; the average count is the rate λ

Figure 1 shows the kind of data the Poisson distribution describes. Events (here, questions from students) arrive at random moments; we cut time into equal intervals (days) and count the events in each. The counts vary from day to day, but their average, the **rate** λ , stays about the same.

The Poisson distribution was named in one line in the [PDF Note](#): a discrete distribution of counts, with parameter λ . This Note teaches it in full:

- what kind of question it answers, and its notation;
- its PMF, with a worked example;
- the shape of its graph and how λ moves it;
- its mean and variance, which are both λ ;
- probabilities of a range of counts;
- its link to the binomial distribution, and where it is used.

The examples follow 365 Data Science.

2. What the Poisson distribution describes

Key point: A Poisson variable counts how often an event happens in one interval; instead of a success probability, we need the average count per interval.

2.1 Counts in an interval

Key point: The binomial distribution needs a number of trials and a probability; the Poisson distribution needs only a rate.

The binomial distribution counts successes in a fixed number n of trials, each with success probability p (see the [Bernoulli and binomial Note](#)). Many counts have no trials to count:

- the questions students send in one day;
- the flashes of a firefly in 10 seconds;
- customers entering a shop in one hour;
- typing errors on one page of a book.

For these we know how **often** the event happens on average, its frequency in a standard interval of time or distance. The **Poisson distribution** turns that average into a probability for every possible count.

For example, a firefly lights up 3 times in 10 seconds on average. The Poisson distribution answers questions such as: how likely is it that it lights up 8 times in 20 seconds?

2.2 Notation

Key point: $Y \sim \text{Po}(\lambda)$ reads “ Y follows a Poisson distribution with rate λ ”.

A distribution is often written as a short name with its parameters in brackets. The symbol $\sim$ reads “follows”:

Notation	Read as	Parameters
$X \sim \text{Bern}(p)$	Bernoulli with success probability p	p
$X \sim B(n, p)$	binomial with n trials and success probability p	n, p
$Y \sim \text{Po}(\lambda)$	Poisson with rate λ	λ

For example, $X \sim B(10, 0.6)$ means 10 trials with success probability 0.6 each, and $Y \sim \text{Po}(4)$ means a count that is 4 on average. The Poisson distribution has a single parameter, λ (lambda), the average number of events per interval.

2.3 The values a Poisson variable can take

Key point: The count starts at 0 and has no upper limit.

A count can never be negative, so the possible values start at 0. Unlike the binomial count, which stops at n , a Poisson count has no cap: in principle the students could send 50 questions in one day. Such large counts are possible, only extremely unlikely.

3. The Poisson PMF

Key point: $P(Y = y) = \frac{\lambda^y e^{-\lambda}}{y!}$, for $y = 0, 1, 2, \dots$

3.1 The example: a spike in questions

Key point: Students usually ask 4 questions a day; yesterday they asked 7. How likely was exactly 7?

Suppose we run an online course on probability. Students usually ask around 4 questions per day, but yesterday they asked 7. To judge whether this spike is unusual, we want the probability of exactly 7 questions in one day.

The three ingredients:

- the interval is one working day;
- the average count per interval is 4, so $\lambda = 4$;
- the count we ask about is 7, so $y = 7$.

3.2 Two pieces of the formula

Key point: $e \approx 2.718$ is a fixed number, and $e^{-\lambda}$ means $1/e^\lambda$.

The formula uses two pieces of notation:

- **Euler's number** e , also called Napier's constant, is a fixed number, $e \approx 2.71828$. It turns up across mathematics, physics and nature; here we only need its value.
- A **negative power** means one over the positive power: $a^{-n} = 1/a^n$. So $e^{-4} = 1/e^4 = 1/54.6 = 0.0183$.

The factorial $y!$ is the product $1 \times 2 \times \dots \times y$, with $0! = 1$ (as in the binomial coefficient of the [Bernoulli and binomial Note](#)).

3.3 Exactly 7 questions

Key point: $P(Y = 7) = 0.060$: about one day in 17 brings exactly 7 questions.

1. **In words:** raise the rate to the power of the count, multiply by e to the power of minus the rate, and divide by the factorial of the count.

2. Formula:

$$P(Y = y) = \frac{\lambda^y e^{-\lambda}}{y!}, \quad y = 0, 1, 2, \dots$$

3. **Example:** with $\lambda = 4$ and $y = 7$:

$$P(Y = 7) = \frac{4^7 e^{-4}}{7!} = \frac{16384 \times 0.0183}{5040} = \frac{300.1}{5040} = 0.0595$$

So there was only about a 6% chance of exactly 7 questions. The bar at 7 in the left panel of Figure 2 has this height.

Extra: A common slip is to read e^{-4} as 0.183. The correct value is 0.0183; with 0.183 the answer would come out as 0.595, ten times too large. The final answer 0.06 needs 0.0183.

Python: `stats.poisson` gives the PMF in one call.

```
import math
from scipy import stats

4 ** 7 * math.exp(-4) / math.factorial(7)    # 0.0595, by hand
# scipy: pmf(y, lambda)
stats.poisson.pmf(7, 4)                      # 0.0595
```

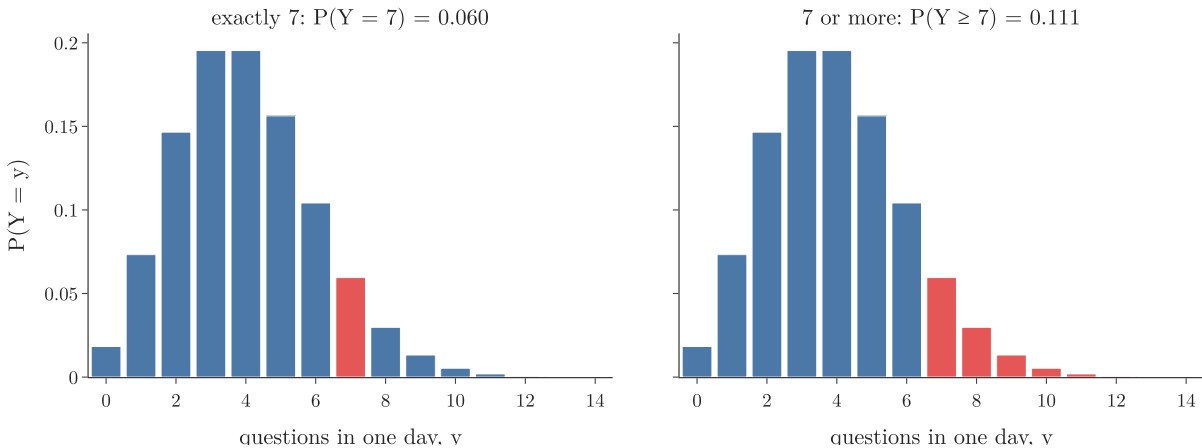


Figure 2: The Poisson PMF with $\lambda = 4$: the bar for exactly 7 questions (left) and the bars for 7 or more (right)

3.4 Scaling the rate to the interval

Key point: λ belongs to one interval length; for a longer or shorter interval, scale it in proportion.

The firefly flashes 3 times in 10 seconds on average, and we ask about 20 seconds. Twice the interval means twice the average, so $\lambda = 3 \times 20/10 = 6$.

1. **In words:** the probability of 8 flashes when 6 are expected.
2. **Formula:**

$$P(Y = 8) = \frac{6^8 e^{-6}}{8!}$$

3. **Example:**

$$P(Y = 8) = \frac{1679616 \times 0.002479}{40320} = \frac{4163.4}{40320} = 0.103$$

Extra: Using $\lambda = 3$ (the rate for 10 seconds) with a 20-second question is a common mistake. The rate and the interval must always match.

4. The graph of the Poisson distribution

Key point: One bar per count, starting at 0 and running on to the right without end; the bars are tallest near λ .

The graph of a Poisson distribution plots each count y against its probability. Figure 2 shows it for $\lambda = 4$:

- it starts at 0, since no event can happen a negative number of times;
- it peaks at 3 and 4 (both have probability 0.195);
- it has a tail to the right that never ends, though after about 12 the bars are too small to see.

The bars add to 1, as in every PMF (see the [PMF Note](#)).

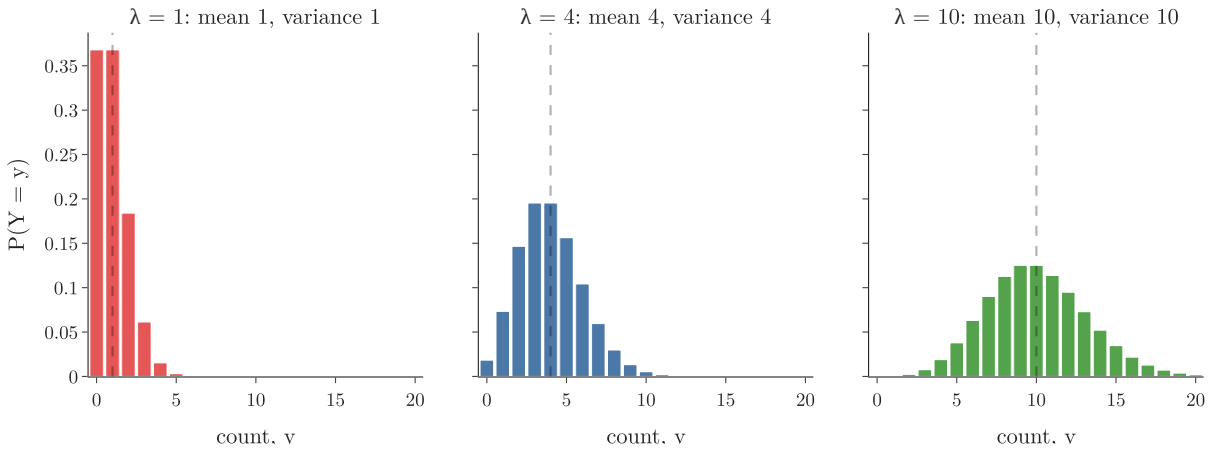


Figure 3: Poisson PMFs for three rates; the dashed line marks the mean λ

Figure 3 shows how λ changes the shape:

- $\lambda = 1$: most days have 0 or 1 events; the distribution is strongly right-skewed (see the [skewness Note](#)).
- $\lambda = 4$: the peak moves right and the bars spread out; a mild right skew remains.

- $\lambda = 10$: the peak sits at 9 and 10, the spread is wider, and the shape is close to a symmetric bell.

Extra: For large λ (about 20 or more) the Poisson distribution is close to a normal distribution with mean λ and variance λ (see the [normal distribution Note](#)). Its skewness is $1/\sqrt{\lambda}$: 1 for $\lambda = 1$, 0.5 for $\lambda = 4$, 0.32 for $\lambda = 10$.

5. Mean and variance

Key point: A Poisson variable has mean λ and variance λ : one number fixes both its centre and its spread.

The expected value is the sum of every value times its probability (see the [expected value Note](#)). Applied to the Poisson PMF, this long sum simplifies to λ . The variance, from the same shortcut formula $E[Y^2] - (E[Y])^2$, is also λ .

1. **In words:** the average count is the rate, and so is the variance; the standard deviation is the square root of the rate.
2. **Formula:**

$$E[Y] = \lambda, \quad \text{Var}(Y) = \lambda, \quad \sigma = \sqrt{\lambda}$$
3. **Example:** for the questions, $\lambda = 4$: on average 4 questions per day, variance 4, standard deviation $\sqrt{4} = 2$. So 7 questions is $(7 - 4)/2 = 1.5$ standard deviations above the mean.

A simulation of 100,000 days with `rng.poisson(lam=4)` gives an average of 3.994 and a variance of 3.987, both close to 4.

Extra: Why the mean is λ . Writing out the sum and cancelling one y against $y!$:

$$E[Y] = \sum_{y=0}^{\infty} y \frac{\lambda^y e^{-\lambda}}{y!} = \lambda \sum_{y=1}^{\infty} \frac{\lambda^{y-1} e^{-\lambda}}{(y-1)!} = \lambda \times 1 = \lambda$$

The remaining sum is the Poisson PMF summed over all counts, which is 1. The same trick applied to $E[Y(Y-1)]$ gives λ^2 , and from it $\text{Var}(Y) = \lambda$.

Extra: Mean equal to variance is a quick check on real count data. If the variance of the counts is much larger than their mean (**overdispersion**), the events are not independent or the rate is not constant, and the Poisson model will underestimate how often extreme counts happen.

6. Probability of a range of counts

Key point: For “at least”, “at most” or “between”, add the probabilities of the single counts in the range.

The counts 0, 1, 2, ... are separate outcomes: one day cannot have both exactly 6 and exactly 7 questions. So the probability of a range is the **sum** of the bars in it, as for any discrete

distribution (see the [PMF Note](#)).

1. **In words:** the probability of 7 or more questions is everything except 0 to 6.
2. **Formula:**

$$P(Y \geq 7) = 1 - P(Y \leq 6) = 1 - \sum_{y=0}^6 \frac{4^y e^{-4}}{y!}$$

3. **Example:** the bars for 0 to 6 are 0.0183, 0.0733, 0.1465, 0.1954, 0.1954, 0.1563 and 0.1042:

$$P(Y \leq 6) = 0.8893, \quad P(Y \geq 7) = 1 - 0.8893 = 0.111$$

The right panel of Figure 2 shows these red bars. A day with 7 or more questions happens about once in 9 days: unusual, but not rare.

More examples with $\lambda = 4$:

Question	Probability
no questions at all, $P(Y = 0)$	0.018
at most 2, $P(Y \leq 2)$	0.238
between 2 and 6, $P(2 \leq Y \leq 6)$	0.798

Python: `cdf` adds the bars up to a count; `sf` adds the bars above it.

```
questions = stats.poisson(4)
questions.sf(6)           # 0.111: P(Y >= 7)
questions.cdf(2)          # 0.238: P(Y <= 2)
questions.cdf(6) - questions.cdf(1) # 0.798: P(2 <= Y <= 6)
```

7. The Poisson distribution as a limit of the binomial

Key point: A binomial count with many trials and a small success probability is close to a Poisson count with $\lambda = np$.

Extra: This whole section is extra material.

A website has 1000 visitors a day, and each visitor buys with probability 0.004. The number of buyers is binomial, $B(1000, 0.004)$, with mean $np = 4$. It is also, very nearly, $Po(4)$.

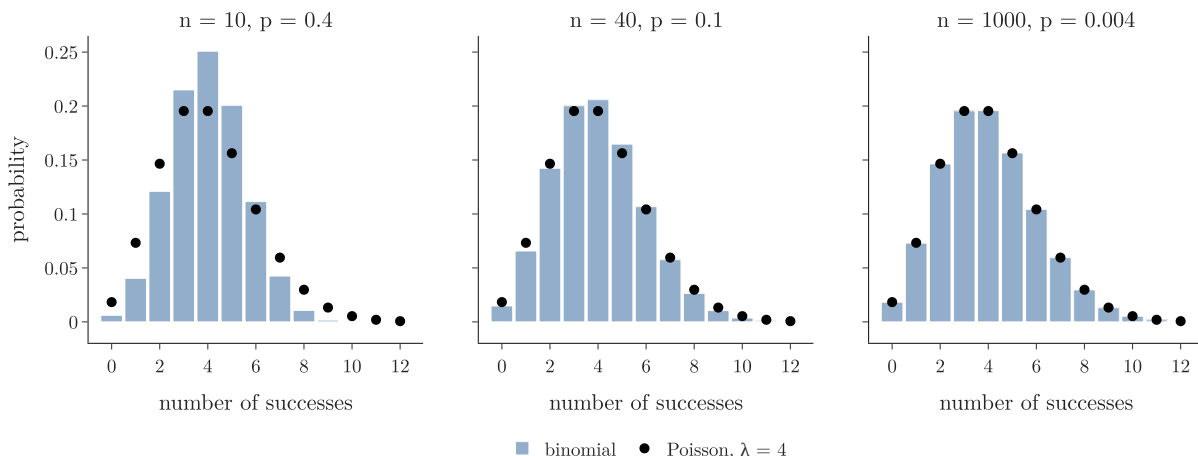


Figure 4: Binomial PMFs with $np = 4$ (bars) against the Poisson PMF with $\lambda = 4$ (dots)

Figure 4 keeps $np = 4$ and lets n grow:

n	p	Largest gap between the two PMFs
10	0.4	0.055
40	0.1	0.011
1000	0.004	0.0004

With 10 trials the binomial is narrower than the Poisson: its variance $np(1 - p) = 2.4$ is below 4. With 1000 trials $1 - p$ is almost 1, the variance $np(1 - p) = 3.98$ is almost λ , and the bars sit on the dots.

This is where the Poisson distribution comes from. Cut a day into many tiny moments; in each, a question arrives or not, with a tiny probability. The count is binomial with huge n and tiny p , and in the limit it becomes Poisson. A rule of thumb: the approximation is good when $n \geq 20$ and $p \leq 0.05$.

8. When a count is Poisson, and where it is used

Key point: Events must happen one at a time, independently, at a constant average rate; then counts of arrivals, calls, clicks or defects are Poisson.

Extra: This whole section is extra material.

A count follows a Poisson distribution when:

1. **Events are independent.** One question does not make the next more or less likely.
2. **The rate is constant.** The average count is the same for every interval of the same length; an exam week with more questions breaks this.

3. **Events happen one at a time.** Two events never happen at exactly the same moment.

Typical Poisson counts:

- customers arriving at a shop or a website per hour;
- calls to a support centre per minute;
- defects per metre of cable, typos per page;
- goals in a football match;
- rare-event counts, such as accidents at a crossing per month.

In machine learning, a target that is a count (bike rentals per hour, insurance claims per year) is often modelled with **Poisson regression**, which predicts λ for each row; scikit-learn has it as **PoissonRegressor**. Counts of words in a document also appear in text models.

9. Summary

	Binomial	Poisson
Counts	successes in n trials	events in one interval
Notation	$B(n, p)$	$\text{Po}(\lambda)$
Parameters	n, p	λ
Values	0 to n	0, 1, 2, ... without limit
PMF	$\binom{n}{y} p^y (1-p)^{n-y}$	$\lambda^y e^{-\lambda} / y!$
Mean, variance	$np, np(1-p)$	λ, λ
Example	2 likes out of 3 viewers, $p = 0.5$: 3/8	7 questions when 4 are usual: 0.060

- The Poisson distribution needs only the average count per interval, λ ; scale λ when the interval changes.
- Its graph starts at 0, peaks near λ and has an endless right tail; a larger λ moves it right and spreads it out.
- Mean and variance are both λ .
- A range of counts has the sum of the single-count probabilities.
- A binomial with large n and small p is approximately Poisson with $\lambda = np$.

10. Key terms

Term	Meaning
Poisson distribution	The distribution of the number of events in a fixed interval, given their average count λ
Rate λ	The average number of events per interval; the only parameter of the Poisson distribution
$Y \sim \text{Po}(\lambda)$	Notation: Y follows a Poisson distribution with rate λ ; likewise $\text{Bern}(p)$ and $B(n, p)$
Euler's number e	A fixed number, about 2.71828, that appears in the Poisson PMF
Negative power	$a^{-n} = 1/a^n$
Overdispersion	Count data whose variance is clearly larger than its mean, a sign that a Poisson model does not fit
Poisson regression	A model that predicts the rate λ of a count target from the features